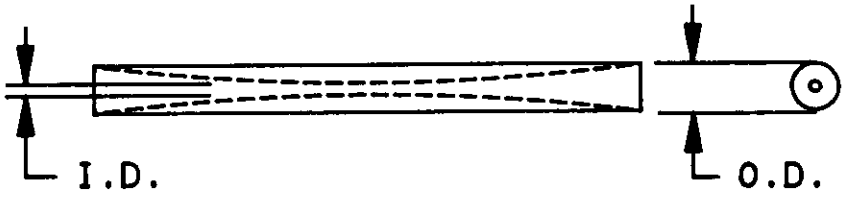
MILITARY SPECIFICATION SHEET

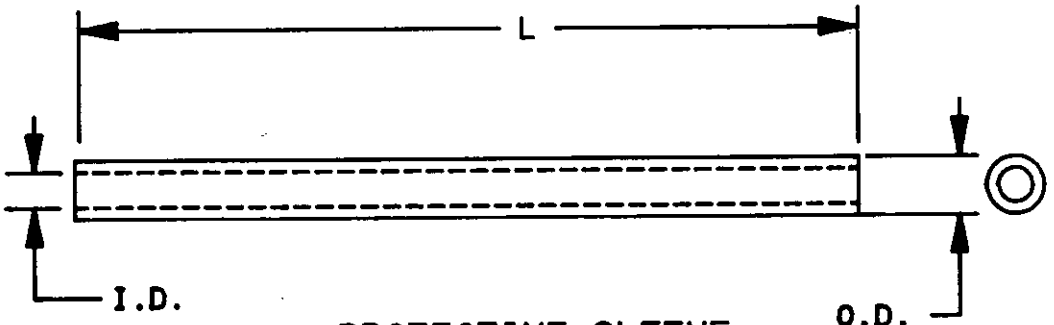
SPLICE, FIBER OPTIC CABLE, FIBER SPLICE (METRIC)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring fiber optic splices described herein shall consist of this specification sheet and the latest issue of DOD-S-24623.



CAPILLARY TUB



PROTECTIVE SLEEVE

Part Number	Item	Dimensions (mm) L	I.D.	O.D.	Mass	Color
D24623/1-1	Capillary Tube	12.70 ±.05 (0.5) (.0020)	0.132 - 0.145 (.0052) (.0057)	0.45 ±.05 (.018) (.0020)	N/A	Transparent
D24623/1-2	Protective Sleeve	34.50 ±.05 (1.358)(.0020)	0.76 - 0.86 (.030) (.034)	1.02 ±.05 (.040) (.0020)	N/A	Transparent

NOTES:
1. Dimensions are in millimeters.
2. Inch dimensions are in parentheses.
3. Inch dimensions are for general information only.

FIGURE 1. Fiber splice construction.

FSC 6060

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REQUIREMENTS:

Materials:

Capillary tube: Quartz.

Protective sleeve: Glass.

Stripping fluid: Swell strip, American Fiber Optics Corp. (AMFOX) or equivalent.

Swell strip 37: Swell strip 37 contains alcohol and some halogenated hydrocarbon and is very effective in stripping optical fibers of their coatings. Its action involves swelling and/or decomposing the polymeric material at room temperature making it easy to remove the latter material. The component chemicals are volatile and flammable, and hazardous to health. Therefore, it should not be heated in any way nor the vapors inhaled.

Cleaning fluid: Acetone.

U V-curable bond (Index matching): U V bond 007 epoxy acrylate, American Fiber Optics Corp. (AMFOX) or equivalent.

U V bond 007: U V bond 007 is an epoxy acrylate formulation for splicing optical fibers. Its optical properties are such that, when cured with ultraviolet light in splicing optical fibers, they have negligible effect in the transmission of light in the fiber. It is chemically stable at room temperature in the absence of any intense or ultraviolet light for a period of 6 months. U V bond 007 is a fairly safe material to handle although, contact with skin should be avoided.

Center wavelength: 850 nm.

Associated tools:

Fiber cleaver.

Mechanical fiber stripper.

Optical performance: Applicable.

Safety labeling:

Capillary tube and protective sleeve - The containers holding the capillary tubes and protective sleeves shall be marked with the following labels:

WARNING

(Protective sleeves) (Capillary tubes) contained herein are made of glass and have sharp edges. Use care when handling to avoid skin penetration or eye damage.

CAUTION

Contents made of glass - fragile. Handle carefully.

Swell strip 37 - The container holding Swell strip 37 shall be marked with the following labels:

WARNING

Contains halogenated hydrocarbons. This is a PROHIBITED material and is NOT allowed onboard submarines.

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CAUTION

Contains alcohol and halogenated hydrocarbons. Do not expose to flame or heat. Do not inhale vapors. Use in well ventilated areas.

U V bond 007 - The container holding U V bond 007 shall be labeled as follows:

CAUTION

Avoid contact with skin.

Sample requirements: Only the physical conformance inspection tests of group I, of table I of DOD-S-24623, and group A inspection tests of table II of DOD-S-24623 shall be performed on the capillary tubes and protective sleeves.

Minimum crush load: Not applicable.

Engineering information: This fiber splice is compatible for use with fiber in DOD-C-85045/1 and with cable splice DOD-S-24623/2.

Nuclear radiation exposure: None.

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - MI
Navy - AS, EC
Air Force - 11, 17, 19, 80, 99
DLA - ES

User activities:
Army - AR
Navy - CG, MC

Agent:
DLA - ES

Preparing activity:
Navy - SH

(Project 6060-0020-1)

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